Okafor Lab Equipment Guide

Revision 3, 10 June 2026. Compiled by Dr. Lucas Ferreira with the equipment owners. Owners and backups are listed in *Who Does What*; this guide says what each instrument is for, how to book it and what goes wrong.

Booking: general rules

- Lab instruments are booked in **MCB-Book** (mcbbook.westbrook.edu, NetID login). You can only book an instrument after its owner has trained you and marked you as trained in MCB-Book.
- Book no more than **14 days ahead**. Cancel slots you will not use. If you have not started within 15 minutes of your slot, the next person may take it.
- **Paper sign-up sheets were retired in January 2026.** A name on a sheet of paper is not a booking, even if an old sheet is still taped to an instrument.
- Every instrument has a logbook next to it. Fill it in each time.
- If something breaks: stop, label it "DO NOT USE" with the date and your name, tell the owner, and post in #equipment. Never try to repair an instrument yourself.
- The confocal and the cryo-EM facility are *not* booked in MCB-Book; see their sections below.

Instruments and areas

AKTA FPLC

Where: cold room 416A (4 °C) *Owner / trainer:* Priya Raghunathan *Booking:* MCB-Book, up to 24 h per booking
Used for: protein purification: affinity, ion exchange and size-exclusion chromatography.

- Filter (0.22 μm) and degas all buffers. Never exceed a column's pressure limit.
- After your run, wash salt out of the system and leave the pumps and columns in 20% ethanol. Log the run count for the shared size-exclusion columns.
- **Common mistakes:** leaving salt in the system (it crystallises in the pumps); leaving a dirty sample loop; propping the cold-room door open. Dress warmly. Never bring dry ice into the cold room (see *Health and Safety*).

Floor ultracentrifuge

Where: 415 *Owner / trainer:* Daniel Osei *Booking:* MCB-Book; overnight runs must be booked by 3 pm that day
Used for: pelleting membranes, density gradients, clearing lysates at more than 100 000 $\times\, g$.

- **The rotor log is mandatory.** For every run record the date, your name, the rotor and its serial number, speed, duration and sample. The rotor log is how we track each rotor's derating and retirement. A run missing from the rotor log is a serious violation; after a second one you lose access.
- **Balance.** Tubes in opposite positions must match to within 0.1 g, or the rotor manual's tolerance if that is stricter. Weigh them on the balance next to the centrifuge. Every position of a swinging-bucket rotor must have its bucket attached, even when a bucket is empty. Fill thin-walled tubes to the specified volume so that they do not collapse.
- Never exceed the rotor's rated speed, or the derated speed written in the rotor log. Wait for the vacuum to be reached before leaving, and for the rotor to stop completely before opening the lid.

- If there is an imbalance alarm or an unusual noise, stop the run, do not open the lid until the rotor has stopped, and tell Daniel.
- Rotors are stored clean, dry and upside down. Aluminium rotors corrode: do not leave salt solutions in them and never clean them with bleach.
- BSL-2 material goes in sealed tubes in a rotor with a sealed lid, and is opened only inside a biosafety cabinet.
- **Common mistakes:** protocols written in rpm instead of $\times g$; wrong tube type for the rotor; forgetting the rotor log.

Multimode plate reader

Where: 415 *Owner / trainer:* Aisha Bello *Booking:* MCB-Book; maximum 3-hour blocks between 9:00 and 18:00 on weekdays; longer kinetic runs overnight
Used for: absorbance, fluorescence and luminescence assays, including kinetics.

- Use the right plate: clear for absorbance, black for fluorescence, white for luminescence.
- **Common mistakes:** leaving data only on the instrument PC (copy it to `okafor-nas` before you leave); leaving the incubator set to 37 °C for the next user.

qPCR machine

Where: 415 *Owner / trainer:* Rafael Castaño *Booking:* MCB-Book, 2-hour blocks
Used for: quantitative PCR and melt curves.

- Seal plates properly and spin them down before loading. **Never open a plate after a run**: amplified products contaminate everything. Discard sealed plates in the biohazard or regular waste, as appropriate.
- **Common mistakes:** poorly sealed plates (evaporation); leaving plates in the machine.

Confocal microscope (shared core facility)

Where: Westbrook Light Microscopy Core, Hartley Building room 210 *Owner / trainer:* core staff (two training sessions, requested on the core website); lab contact Hannah Lindqvist *Booking:* the core's own CoreSched system, not MCB-Book
Used for: fixed-cell and live-cell fluorescence imaging.

- Use is charged hourly to the lab. Ask Marisol for the chartstring before your first booking. Cancel at least 24 h ahead or the lab is charged for the slot.
- BSL-2 samples must be fixed, or sealed in the core's approved chambers.
- **Common mistakes:** skipping the shutdown sequence (lasers left on); booking long tile scans in the core's prime time.

Biosafety cabinets (BSC-1, BSC-2, BSC-3)

Where: tissue culture room 413 *Owner / trainer:* Mei-Ling Zhou (TC sign-off) *Booking:* BSC-1 and BSC-2 in MCB-Book in 2-hour blocks; BSC-3 is the quarantine cabinet for cell lines not yet tested for mycoplasma
Used for: all open manipulation of cells and other BSL-2 material.

- Check the certification sticker (certified every year) and the airflow indicator. Let the cabinet run for 5 minutes before starting. Work at least 10 cm inside the front grille and never block the grilles. No open flames.
- Wipe down with 70% ethanol before and after work. The UV lamp is not a substitute for wiping down. Full BSL-2 rules are in *Health and Safety*.

- **Common mistakes:** sash above the marked height; clutter blocking the airflow; leaving the aspirator flask more than half full.

CO_2 incubators (Inc-1 to Inc-4)

Where: 413 *Owner / trainer:* Mei-Ling Zhou *Booking:* no booking; shelves are allocated by name
Used for: growing mammalian cells at 37 °C, 5% CO_2.

- Label every flask or dish with your initials, cell line, passage number and date. Unlabelled vessels are discarded on Fridays. Inc-4 is the quarantine incubator for lines not yet tested for mycoplasma.
- **Common mistakes:** leaving the door open; overcrowding shelves; ignoring the "CO_2 low" alarm (see *After Hours and Special Materials*).

Autoclave (shared with the Nakamura Lab)

Where: Keller 420 *Owner / trainer:* Marisol Vega (15-minute training) *Booking:* no booking; waste runs follow the chore rota
Used for: sterilising biohazard waste, glassware, media components and solutions.

- **Never autoclave sealed containers**: loosen caps. Never autoclave bleach, solvents or other chemicals.
- Wear heat-resistant gloves and a face shield when unloading liquids. Wait until the chamber pressure reads zero, open the door a crack, and let liquids stand 10 minutes before removing them.
- Log every run. **Common mistakes:** tightly closed bottles (they can burst); overfilled biohazard bags; wrong cycle (liquid vs. dry).

-80 °C freezers (A, B, C) and shelf allocation

Where: 416 *Owner / trainer:* Marisol Vega *Booking:* no booking
Used for: long-term storage of samples, purified proteins, competent cells and enzymes.

Shelf	Freezer A	Freezer B	Freezer C (backup)
1	Common lab stocks (competent cells, enzymes)	Purified protein archive	Kept empty for emergencies
2	Priya Raghunathan, Lucas Ferreira	Mouse tissue (animal protocol personnel only)	Kept empty for emergencies
3	Hannah Lindqvist, Daniel Osei	Undergraduates and rotation students (one labelled box each)	Overflow, by arrangement with Marisol
4	Mei-Ling Zhou, Rafael Castaño, Aisha Bello	Shared antibody overflow	Overflow, by arrangement with Marisol
5	Tomasz Brenner	Spare; ask Marisol	Overflow, by arrangement with Marisol

- Every box is labelled with your name and the date and entered in the freezer inventory (`okafor-nas/admin/inventory/freezers.xlsx`). Keep the door open as briefly as possible. Never put boxes on someone else's shelf.
- Rotation students: when your rotation ends, either transfer your box to your rotation mentor's shelf or discard it, and update the inventory.
- What to do if a freezer alarms is in *After Hours and Special Materials*.

Liquid nitrogen dewars

Where: 416 *Owner / trainer:* Tomasz Brenner (Cryogen Safety course plus hands-on sign-off) *Booking:* no booking

Used for: cryostorage of cell lines. Dewar 1 holds master stocks (access with Tomasz only), Dewar 2 holds working stocks, and a 180-L supply dewar is used for filling.

- PPE and filling rules are in *After Hours and Special Materials*. An oxygen monitor on the wall of 416 alarms if oxygen falls.

Gel and blot imager

Where: 415 *Owner / trainer:* Lucas Ferreira *Booking:* no booking; first come, first served; 30 minutes maximum when others are waiting
Used for: chemiluminescent and fluorescent western blots, stained gels.

- Use the tray. Wipe it afterwards. Save the uncropped raw image file to `okafor-nas` (see *Writing and Notebook Style*).
- **Common mistakes:** saturated exposures (turn on the saturation overlay); exporting only cropped JPEGs.

NanoDrop spectrophotometer

Where: 415 *Owner / trainer:* Aisha Bello *Booking:* no booking
Used for: quick nucleic acid and protein concentration measurements.

- Blank with the same buffer your sample is in. Clean both pedestals with water and a dry lab wipe after use. **Common mistake:** a dried sample left on the pedestal.

Probe sonicator

Where: sound enclosure in 412B *Owner / trainer:* Priya Raghunathan *Booking:* MCB-Book, 1-hour blocks
Used for: lysing cells and shearing DNA.

- **Hearing protection is required** for everyone in 412B while the sonicator runs (ear defenders hang by the door). Keep the enclosure door closed.
- Keep samples on ice and use pulse mode. Never run the probe in air. Clean the tip with water and then 70% ethanol.

Cryo-EM sample preparation (facility)

Where: Westbrook Cryo-EM Facility, Sato Center room 150 *Owner / trainer:* facility staff; lab contact Lucas Ferreira *Booking:* through the facility's own system; talk to Lucas first
Used for: grid preparation (plunge freezing) and screening.

- The facility trains and certifies you on its plunge freezer and on handling cryogens there. Its rules apply inside the facility.